

Counterexamples to a conjecture on one-sided restricted sumsets

September 8, 2026

Abstract

For finite integer sets A, B and a forbidden relation $R \subseteq A \times B$, let $A +_R B$ be the set of sums represented by pairs outside R . Ouyang conjectured that, when $|B| \leq |A|$ and every vertex of B has forbidden degree at most Δ , one has $|A +_R B| \geq |A| + |B| - 1 - \lfloor 5\Delta/2 \rfloor$. We give explicit counterexamples for every $\Delta = 3k$, $k \geq 1$. For every $N \geq 14k$, our sets have cardinality N and restricted sumset cardinality $2N - 8k - 1$, falling below the conjectured bound by $\lceil k/2 \rceil$. Consequently a universal linear loss coefficient must be at least $8/3$. For $\Delta = 3$ and equal summand sizes $N \geq 14$, the construction attains the known lower bound $2N - 9$, and hence determines the exact minimum.

1 Introduction

Let $A, B \subseteq \mathbb{Z}$ be finite, nonempty sets and let $R \subseteq A \times B$. The restricted sumset associated with the forbidden relation R is

$$A +_R B = \{a + b : a \in A, b \in B, (a, b) \notin R\}.$$

For $b \in B$, put $d_R(b) = |\{a \in A : (a, b) \in R\}|$. We consider the one-sided condition $d_R(b) \leq \Delta$ for all $b \in B$. Degrees on A are unrestricted.

Ouyang proved that, if $|B| \leq |A|$ and the one-sided degree condition holds, then

$$|A +_R B| \geq |A| + |B| - 3\Delta \tag{1}$$

[1, Theorem 1.5(i)]. Motivated by a construction in [1, Theorem 1.9], Ouyang proposed the stronger inequality

$$|A +_R B| \geq |A| + |B| - 1 - \left\lfloor \frac{5\Delta}{2} \right\rfloor \tag{2}$$

[1, Conjecture 1.10]. All numbered references to that paper refer to its September 4, 2025 author version, arXiv:2503.09121v3.

We disprove (2) by a family whose forbidden degree is unbounded. The construction uses an interval with three holes as A , an interval as B , and four intervals of deleted sums. The holes near the endpoints of A control the degree when both interior intervals of deleted sums meet the same translate of A . Exact intersection counts show that the resulting degree is at most $3k$, while the loss from $|A| + |B| - 1$ is $8k$.

Throughout, $[x, y]$ denotes the integer interval $\{x, x + 1, \dots, y\}$ for integers $x \leq y$.

Theorem 1. *Let k, N, U be integers satisfying*

$$k \geq 1, \quad N \geq 14k, \quad 8k \leq U \leq N - 6k. \tag{3}$$

Define

$$A = [0, N + 4k - 1] \setminus ([k, 2k - 1] \cup [U, U + 2k - 1] \cup [N + 2k, N + 3k - 1]),$$

$$B = [0, N - 1],$$

and put

$$\begin{aligned} D_0 &= [0, 4k - 1], & D_1 &= [U, U + 2k - 1], \\ D_2 &= [U + N - 1, U + N + 2k - 2], & D_3 &= [2N - 1, 2N + 4k - 2]. \end{aligned}$$

Let $D = D_0 \cup D_1 \cup D_2 \cup D_3$ and

$$R = \{(a, b) \in A \times B : a + b \in D\}.$$

Then

$$\begin{aligned} |A| &= |B| = N, & \max_{b \in B} d_R(b) &= 3k, \\ A +_R B &= [0, 2N + 4k - 2] \setminus D, & |A +_R B| &= 2N - 8k - 1. \end{aligned}$$

Corollary 2. *Conjecture 1.10 of [1] is false. For each $k \geq 1$, Theorem 1 gives examples with $\Delta = 3k$ whose restricted sumset has $\lceil k/2 \rceil$ fewer elements than the proposed lower bound.*

Proof. The proposed lower bound is $2N - 1 - \lfloor 15k/2 \rfloor$, whereas the actual cardinality is $2N - 8k - 1$. Their difference is

$$8k - \left\lfloor \frac{15k}{2} \right\rfloor = \left\lceil \frac{k}{2} \right\rceil > 0.$$

□

Corollary 3. *Suppose real constants c, K , independent of A, B, Δ , satisfy*

$$|A +_R B| \geq |A| + |B| - 1 - c\Delta - K$$

for all finite, nonempty $A, B \subseteq \mathbb{Z}$ with $|B| \leq |A|$ and all relations of degree at most Δ on B . Then $c \geq 8/3$.

Proof. Apply Theorem 1 with $N = 14k$ and $U = 8k$. The claimed universal bound implies $8k \leq 3ck + K$. Divide by k and let k tend to infinity. □

2 Proof of the construction

Proof of Theorem 1. The three holes in A are disjoint and have total cardinality $4k$. Hence $|A| = N$, and clearly $|B| = N$. Both endpoints 0 and $N + 4k - 1$ belong to A . The largest difference between consecutive elements of A is $2k + 1 \leq N$. Thus the intervals $a + B = [a, a + N - 1]$, as a runs through A in increasing order, overlap or are adjacent. It follows that

$$A + B = [0, 2N + 4k - 2]. \tag{4}$$

The parameter inequalities imply that D_0, D_1, D_2, D_3 are disjoint, lie in this interval, and have lengths $4k, 2k, 2k, 4k$, respectively. Consequently $|D| = 12k$.

It remains to prove the degree assertion. For $b \in B$, define

$$d_i(b) = |(A + b) \cap D_i| \quad (0 \leq i \leq 3), \quad d(b) = \sum_{i=0}^3 d_i(b) = d_R(b).$$

The interval D_0 can contribute only when $0 \leq b \leq 4k - 1$, and D_3 can contribute only when $N - 4k \leq b \leq N - 1$. The two interior intervals can both contribute only when

$$U - 4k \leq b \leq U + 2k - 1.$$

Indeed, meeting D_2 requires $N + 4k - 1 + b \geq U + N - 1$, while meeting D_1 requires $b \leq U + 2k - 1$. The three ranges

$$[0, 4k - 1], \quad [U - 4k, U + 2k - 1], \quad [N - 4k, N - 1] \quad (5)$$

are pairwise disjoint: $U - 4k \geq 4k$ and $U + 2k - 1 \leq N - 4k - 1$. Outside their union, neither endpoint interval contributes and at most one interior interval contributes. Its length is $2k$, so $d(b) \leq 2k$ there.

The left endpoint range. Suppose $0 \leq b \leq 4k - 1$. Only D_0 and D_1 contribute. The first count is the number of elements in $[0, 4k - 1 - b]$ after removing $[k, 2k - 1]$. The second is the number of elements of A in $[U - b, U + 2k - 1 - b]$. This interval can meet the central hole $[U, U + 2k - 1]$ but no other hole. Direct counting gives

range of b	$d_0(b)$	$d_1(b)$	$d(b)$
$0 \leq b \leq 2k - 1$	$3k - b$	b	$3k$
$2k \leq b \leq 3k - 1$	k	$2k$	$3k$
$3k \leq b \leq 4k - 1$	$4k - b$	$2k$	$6k - b$

(6)

Every entry in the last column is at most $3k$.

The middle range. Write $b = U - 4k + t$ with $0 \leq t \leq 6k - 1$. Only D_1 and D_2 contribute. To compute $d_1(b)$, intersect A with

$$D_1 - b = [4k - t, 6k - 1 - t].$$

This interval meets only the left endpoint of A and the hole $[k, 2k - 1]$; its upper endpoint is less than U . For the second count,

$$D_2 - b = [N + 4k - 1 - t, N + 6k - 2 - t].$$

Set $L = N + 4k - 1$. Reflection $x \mapsto L - x$ takes this interval to $[t - 2k + 1, t]$. The reflected set $L - A$ has the endpoint hole $[k, 2k - 1]$ and its central hole begins at $N + 2k - U \geq 8k$. Since $t \leq 6k - 1$, no other hole contributes to this count. The exact intersection sizes are therefore

range of t	$d_1(b)$	$d_2(b)$	$d(b)$
$0 \leq t \leq k - 1$	$2k$	$t + 1$	$2k + t + 1$
$k \leq t \leq 2k - 1$	$2k$	k	$3k$
$2k \leq t \leq 3k - 1$	$4k - t$	k	$5k - t$
$3k \leq t \leq 4k - 1$	k	$t - 2k + 1$	$t - k + 1$
$4k \leq t \leq 5k - 1$	k	$2k$	$3k$
$5k \leq t \leq 6k - 1$	$6k - t$	$2k$	$8k - t$

(7)

Again every entry in the last column is at most $3k$.

The right endpoint range. The left endpoint calculation also handles this range by reflection. To make the parameter change explicit, write A_U, D_U for the constructed sets. With

$$U' = N + 2k - U, \quad M = 2N + 4k - 2,$$

we have

$$L - A_U = A_{U'}, \quad M - D_U = D_{U'}, \quad (N - 1) - B = B.$$

Moreover, $8k \leq U' \leq N - 6k$. The map

$$(a, b) \mapsto (L - a, N - 1 - b)$$

therefore carries the forbidden relation for U to the forbidden relation for U' . For $b \in [N - 4k, N - 1]$, the new coordinate $N - 1 - b$ lies in $[0, 4k - 1]$, where (6) applies. This proves $d(b) \leq 3k$ in the right endpoint range.

Together these calculations establish $d_R(b) \leq 3k$ for every $b \in B$. Equality holds at $b = 0$ by (6). Finally, by the definition of R , every representation of a sum in D is forbidden and every representation of a sum outside D is permitted. Using (4), we obtain

$$A +_R B = (A + B) \setminus D = [0, 2N + 4k - 2] \setminus D.$$

Its cardinality is $(2N + 4k - 1) - 12k = 2N - 8k - 1$, as required. $\square$

3 Exact sharpness at degree three

For completeness, we recall the elementary lower-bound argument underlying (1). This is the argument of [1, Theorem 1.5(i)].

Proposition 4 (Ouyang). *Let $A, B \subseteq \mathbb{Z}$ be finite, nonempty sets, with $|B| \leq |A|$, and let $\Delta \geq 1$ be an integer. If $R \subseteq A \times B$ has degree at most Δ on B , then $|A +_R B| \geq |A| + |B| - 3\Delta$.*

Proof. Write $A = \{a_1 < \dots < a_m\}$ and $B = \{b_1 < \dots < b_n\}$. If a_1 has forbidden degree at most Δ , the strictly increasing chain of sums along

$$(a_1, b_1), \dots, (a_1, b_n), (a_2, b_n), \dots, (a_m, b_n)$$

has $m+n-1$ elements and at most 2Δ forbidden pairs. It supplies at least $m+n-1-2\Delta \geq m+n-3\Delta$ permitted sums.

Otherwise, a_1 has degree at least $\Delta + 1$. Since

$$\sum_{a \in A} d_R(a) = \sum_{b \in B} d_R(b) \leq n\Delta \leq m\Delta,$$

there is some a_j with $j \geq 2$ whose degree is at most $\Delta - 1$. Consider the chain along the first column up to a_j , then along the row of a_j , then along the last column:

$$(a_1, b_1), \dots, (a_j, b_1), (a_j, b_2), \dots, (a_j, b_n), (a_{j+1}, b_n), \dots, (a_m, b_n).$$

It contains $m+n-1$ strictly increasing sums. The number of forbidden pairs is at most $\Delta + (\Delta - 1) + \Delta = 3\Delta - 1$, since the two columns have degree at most Δ . Thus at least $m+n-3\Delta$ sums remain. $\square$

Corollary 5. *For every integer $N \geq 14$,*

$$\min |A +_R B| = 2N - 9,$$

where the minimum is over finite integer sets A, B of cardinality N and relations $R \subseteq A \times B$ of degree at most three on B .

Proof. Proposition 4 gives the lower bound. Theorem 1, with $k = 1$ and any $8 \leq U \leq N - 6$, attains it. $\square$

For example, take $k = 1$, $N = 20$, and $U = 14$. Then

$$\begin{aligned} A &= \{0\} \cup [2, 13] \cup [16, 21] \cup \{23\}, \\ B &= [0, 19], \\ D &= \{0, 1, 2, 3, 14, 15, 33, 34, 39, 40, 41, 42\}. \end{aligned}$$

The degree sequence on B , in increasing order, is

$$(3, 3, 3, 3, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3).$$

The restricted sumset has 31 elements, whereas (2) requires 32.

The construction leaves a gap in the possible universal leading coefficient: Corollary 3 requires at least $8/3$, while (1) supplies 3. We do not determine that coefficient, classify the equality cases at degree three, or assert a corresponding counterexample under a degree restriction on both sides.

References

- [1] Minghui Ouyang, *On restricted sumsets with bounded degree relations*, *Mathematika* **71** (2025), no. 4, e70045. doi:10.1112/mtk.70045. Author version: arXiv:2503.09121v3, September 4, 2025.